

Metal-insulator transition at low densities in a two-dimensional low-bandgap semiconductor

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Metal-insulator transitions in low band-gap systems emerge from the interplay of interactions, disorder, and band structure on comparable energy scales. Platinum diselenide, with its thickness-dependent electronic structure, provides an ideal platform to explore this regime, as the five-layer limit lies close to the semimetallic boundary while retaining a small bandgap (~0.1 eV). Here we show that hexagonal boron nitride-encapsulated five-layer PtSe₂ devices with embedded ultra-flat, pre-patterned platinum contacts exhibit high electronic quality, with mobilities up to 1,640 cm²V⁻¹s⁻¹ and contact resistances as low as 1.31 kΩ-μm at 1.5 K. These characteristics enable a gate-tunable metal-insulator transition at carrier densities down to 2.75×10^{10} cm⁻². Transport in this regime is dominated by electron-electron interactions, as supported by finite-temperature scaling. Below 50 K, however, an additional disorder-related energy scale emerges, consistent with percolation. These results demonstrate that disorder remains relevant even in high-mobility devices, challenging a purely interaction-driven transition in two-dimensional systems.

Phase transitions in condensed-matter systems are fundamental for understanding the collective dynamics of quantum many-body systems. In two-dimensional (2D) materials, metal-insulator transitions (MITs) provide a particularly compelling platform for exploring these phenomena, extending beyond single-electron models to encompass complex many-body interactions^{1,2}. In this context, the zero-temperature quantum phase transition (QPT) model emphasizes the role of strong carrier-carrier interactions in driving MITs, while the percolation model focuses on how spatial disorder disrupts conductive pathways. Both models contribute to our understanding of

MITs, but the interplay between these mechanisms in 2D materials remains not fully understood.

Discrepancies in observed transitions in 2D semiconductors highlight the complexities involved, compounded by sample-to-sample variations and inconsistencies. For instance, some studies on MoS₂ suggest that interaction-driven MITs are predominant^{3,4}, while others attribute MITs in MoS₂ transistors to percolation mechanisms^{5,6}. These discrepancies underscore the challenges in understanding MITs in 2D systems and have led to debates about the feasibility of observing quantum phases like the Wigner crystal (WC) in 2D systems due to

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inherent disorder⁷. To investigate MIT phenomena and mechanisms effectively in 2D systems, it is crucial to fabricate high-quality devices.

Recent advancements in high-quality monolayer WSe₂ devices⁸, enabled by innovative “charge-transfer contacts” and superior crystal growth techniques, have demonstrated MITs at exceptionally low carrier densities ($\sim 1.6 \times 10^{11} \text{ cm}^{-2}$). These studies report that the observed transitions are consistent with a QPT, leading to the formation of a WC in the insulating phase. Conversely, theoretical models suggest that Coulomb disorder in 2D systems might prevent the stabilization of a WC, potentially resulting in a localized Anderson insulator instead⁷. This theoretical perspective underscores the complexity of disorder’s role in 2D materials and highlights the subtle interplay between disorder and interactions in MITs, which continues to be an area of active investigation in both experimental and theoretical studies.

In this context, platinum diselenide (PtSe₂), a relatively unexplored transition metal dichalcogenide (TMD)⁹, emerges as a promising candidate for studying MITs. In comparison to other layered TMDs, this material is characterized by a rather strong coupling between the electronic states of individual monolayer counterparts, which results in strong dependence of its electronic properties on thickness accompanied by strong electron-electron interactions and spin-orbit coupling^{9–13}. Notably, PtSe₂ undergoes a thickness-dependent transition from semimetal for the bulk material to semiconductor with a relatively large gap for its monolayer. Theoretical predictions indicate the potential for high field-effect mobilities¹⁴; however, experimental results with directly evaporated contacts have frequently reported much lower mobilities^{15,16}. Nevertheless, with the fabrication of high-quality devices, PtSe₂ offers the potential for studying MITs in detail.

In this study, we experimentally characterize high-quality, hexagonal boron nitride (hBN)-encapsulated PtSe₂ devices with embedded, ultra-flat, pre-patterned Pt contacts. We focus on 4-layer (4L) and

5-layer (5L) PtSe₂ field-effect transistors (FETs). Among the semi-conducting forms, 5L and 4L PtSe₂ exhibit the smallest bandgaps, approximately 0.1 eV and 0.26 eV, respectively. The formation of transparent ohmic contacts is observed at backgate voltages above 7.5 V down to ~ 1.5 K, with contact resistances as low as $\sim 1.31 \text{ k}\Omega \mu\text{m}$, and high 2-terminal field-effect mobilities up to $\sim 650 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and 4-terminal mobilities up to $\sim 1640 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. These devices enable the observation of a gate-tuned MIT in 5L at ultralow carrier densities ($\sim 2.75 \times 10^{10} \text{ cm}^{-2}$), and in 4L at slightly higher densities ($\sim 5.88 \times 10^{10} \text{ cm}^{-2}$). At these densities, a large interaction parameter of $r_s \sim 20.6$ is obtained, indicating strong carrier-carrier interactions at higher temperatures (> 50 K), consistent with an interaction-dominated MIT, which is characterized using finite-temperature scaling. However, at temperatures below 50 K, charge transport enters the charge-impurity-scattering regime, inducing disorder and suggesting a crossover to a percolation-driven MIT. Our findings underscore the critical interplay between disorder and interactions in shaping MITs in 2D materials.

Results

Our device architecture is depicted schematically in Fig. 1a. Fabrication begins with the mechanical exfoliation and identification of ~ 30 nm thick bottom hBN flakes on doped Si wafers with a 300 nm SiO₂ layer, which serves as the backgate (Fig. 1b). Standard e-beam lithography is then used to define a Hall-bar contact geometry, followed by etching the exposed regions by ~ 15 nm using deep reactive ion etching. Ti/Pt (2/14 nm) electrodes are deposited by e-beam evaporation to form the pre-patterned contacts. Afterwards, the device undergoes an initial annealing step under high-vacuum conditions ($\sim 1 \times 10^{-6}$ mbar) at 340 °C for 6 h to remove fabrication-related residues. Next, a bilayer of top hBN and PtSe₂—each picked-up from SiO₂ substrates using the

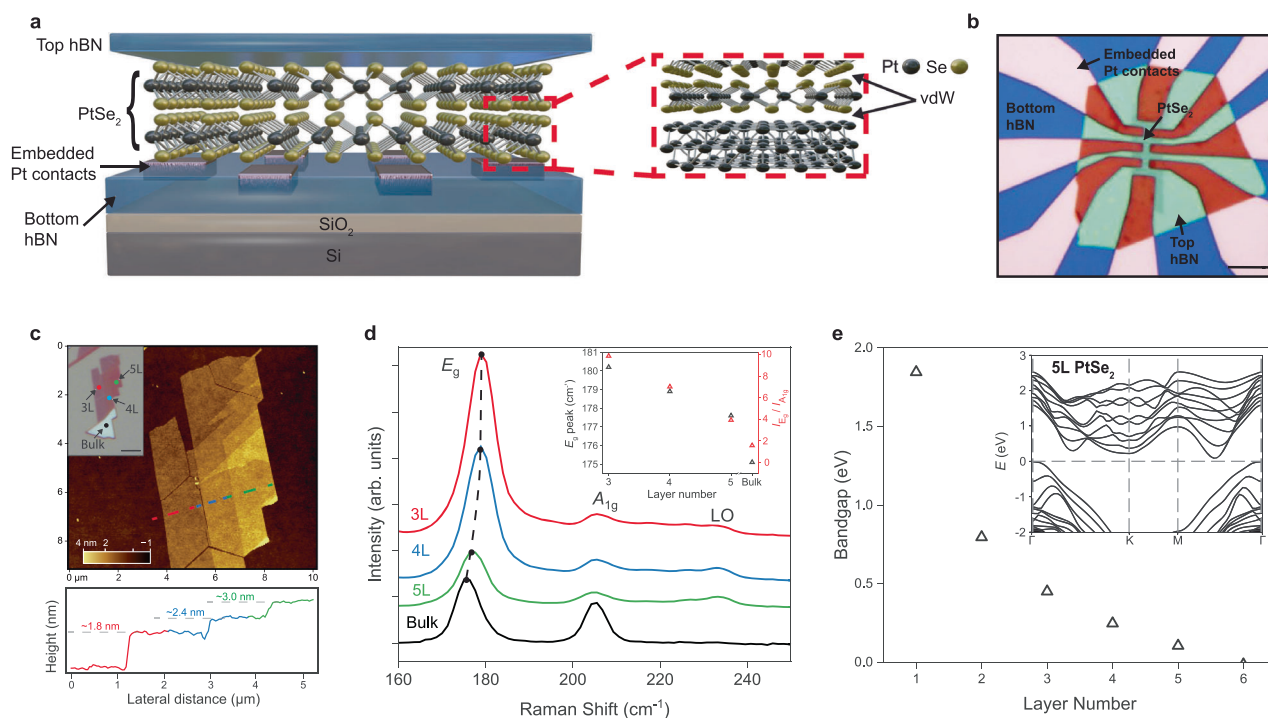


Fig. 1 | Device architecture and characterization. **a** Schematic of PtSe₂ device with a cross-sectional view of the Pt/PtSe₂ interface, highlighting the van der Waals (vdW) gap. Crystal structure models are constructed using VESTA⁴⁷. **b** Optical micrograph of 5L PtSe₂ device, the contrast has been adapted to distinguish the different layers. The scale bar is equal to 5 μm . **c** Top panel: AFM image of PtSe₂ flake with varying thickness from 3L to 5L. Bottom panel: AFM height measurement for the thickness of 3L to 5L PtSe₂. Inset: Optical micrograph of PtSe₂ flake measured

by AFM, the scale bar is equal to 3 μm . **d** Normalized Raman spectra measured for the different regions of the flake in c) with corresponding colors to thickness. A vertical offset has been applied to distinguish the different curves; the dashed line illustrates the E_g peak shift as thickness changes. Inset: Dependence of E_g peak shift (black) and I_{Eg}/I_{A1g} (red) on PtSe₂ thickness. **e** PtSe₂ bandgap against layer number obtained by hybrid functional DFT calculations. Inset: the band structure of 5L PtSe₂.

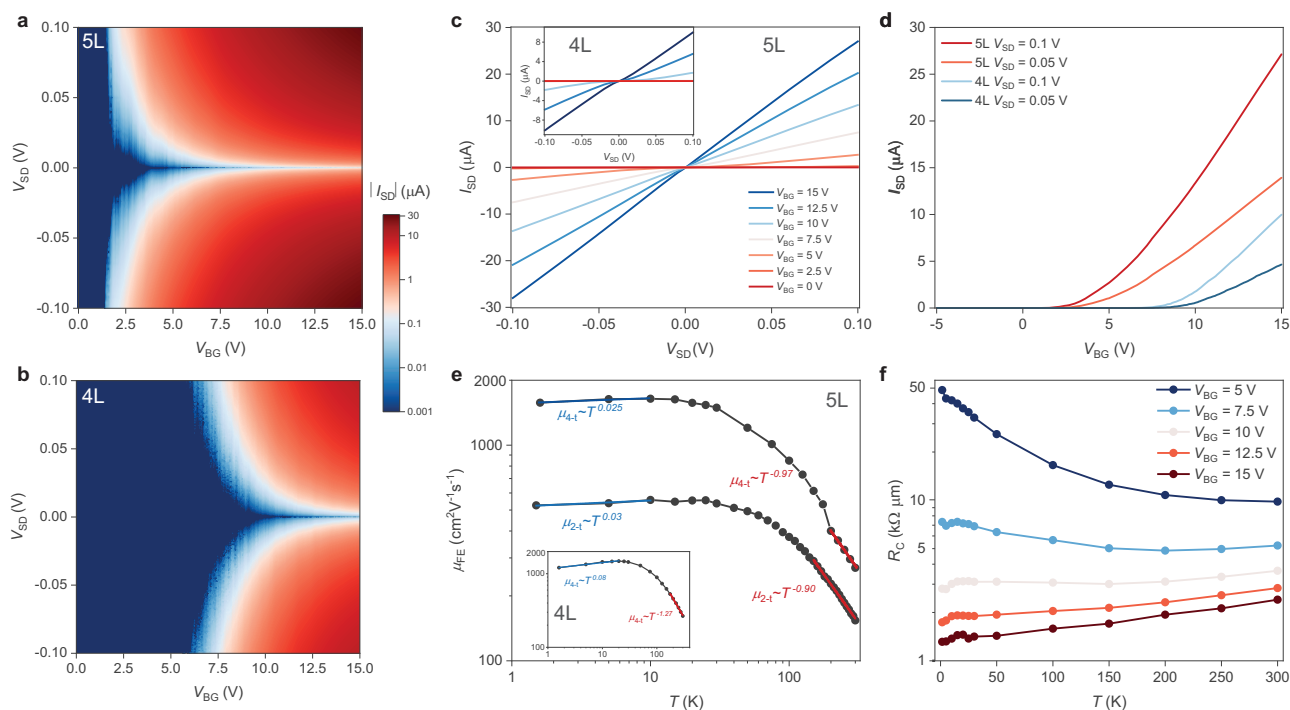


Fig. 2 | Electrical transport characteristics of 4 L and 5 L PtSe₂ devices. a, b 2D color maps of I_{SD} modulation with V_{SD} and V_{BG} on a log-scale for the 5 L (a) and 4 L (b) devices, measured at $T = 1.5$ K in a two-terminal configuration. Both color maps share the same scale, emphasizing the enhanced conduction in the 5 L device compared to the 4 L device. **c** $I - V_{SD}$ characteristics of the 5 L device at various V_{BG} , showing strong linearity and confirming ohmic behavior at $T = 1.5$ K. Inset: $I - V_{SD}$ ohmic characteristics of the 4 L device. **d** $I - V_{BG}$ field-effect transistor (FET) transfer curves at different V_{SD} for both 4 L and 5 L devices at $T = 1.5$ K. **e** Temperature

dependence of 2-t and 4-t electron field-effect mobility for the 5 L device. Power-law fits ($\mu \sim T^\gamma$) indicate distinct regimes governed by phonon and charge impurity scattering. Inset: 4-t field-effect mobility as a function of temperature for the 4 L device, with similar power-law fittings. We note that 4-terminal measurements were conducted under an AC current bias of $I_{SD}^{AC} = 1 \mu A$. **f** R_C versus temperature for different V_{BG} values in the 5 L device. At higher V_{BG} values, contact resistance increases with increasing temperature as expected for ohmic contacts.

polycarbonate-based dry transfer method¹⁷ is transferred onto the pre-patterned Pt contacts. By embedding the Pt contacts within the bottom hBN dielectric, a nearly flat surface is achieved for PtSe₂ deposition, improving the interface quality between hBN and PtSe₂ and preventing strain, thus enhancing overall device performance. To complete the device, 2 nm Ti/100 nm Au contacts are formed over the unexposed Pt contact regions using additional e-beam lithography and evaporation. The fabrication ends with a final annealing step, performed under the same conditions, which further optimizes contact quality and removes residual impurities.

As the thickness of PtSe₂ decreases below six layers, its band structure transitions from semimetallic to semiconducting, with the bandgap increasing from 105 meV in the 5-layer (5L) structure to 1.85 eV in the monolayer, as confirmed by our hybrid functional DFT calculations (Fig. 1e). For our study, we intentionally selected 4 L and 5 L flakes as they exhibit the smallest band gaps just below the semimetallic response. The thinnest flakes tend to exhibit larger Schottky barriers and lower mobilities due to their higher band gaps, whereas 4 L and 5 L flakes provide higher mobility, reduced defect density, and lower contact resistance, making them ideal candidates for investigating gate-voltage-driven MIT. Since the electronic properties of PtSe₂ are highly dependent on layer thickness, here we used both AFM and Raman spectroscopy to precisely characterize the number of layers. An example AFM height measurement of 3 L, 4 L, and 5 L PtSe₂ is shown in Fig. 1c, alongside corresponding Raman spectra in Fig. 1d. As the thickness decreases to 3 layers, the E_g peak shifts from 175 cm⁻¹ in the bulk to 180.2 cm⁻¹, while the ratio of the intensity of E_g to the intensity of A_{1g} increases from 1.6 (bulk) to 9.8 (3L) (Fig. 1d - inset). These monotonic shifts show the effectiveness of Raman spectroscopy for identifying PtSe₂ flakes of semiconducting thickness. Utilizing both

characterization techniques, we fabricated devices based on 4 L and 5 L PtSe₂ flakes for this study. Unless otherwise stated, the results presented here are from a representative 5 L PtSe₂ device, shown optically in Fig. 1b.

Figure 2a, b show 2D color mappings at $T = 1.5$ K of source-drain current (I_{SD}) as a function of V_{SD} and V_{BG} for the 4 L and 5 L devices, respectively, obtained through two-terminal measurements. The transition from 4 L to 5 L results in a notable shift in the threshold voltage V_{TH} from 5.5 V to 1.4 V, reflecting the smaller bandgap in 5 L PtSe₂¹⁸. To provide a direct comparison of the conductivity, we evaluate the drain current at the same relative gate voltage, defined as $\Delta V_{BG} = V_{BG} - V_{TH}$. For the 4 L device, $V_{BG} = 15$ V corresponds to $\Delta V_{BG} = 9.5$ V. At a similar doping level in the 5 L device ($V_{BG} \sim 11$ V), the drain current is $I_{SD} \sim 15 \mu A$ under $V_{SD} = 0.1$ V, compared to $I_{SD} \sim 10 \mu A$ for the 4 L device. This comparison reveals a 50 % increase in conductivity for the 5 L device, supporting the observed increase in density of states and confirming the expected relationship between bandgap and device performance.

Vertical and horizontal line segments of the 2D color maps in Fig. 2a, b are presented in Fig. 2c, d. The $I - V_{SD}$ characteristics of the contacts in the 5 L device, shown in Fig. 2c, exhibit linearity for $V_{BG} > 7.5$ V at $T = 1.5$ K, indicating the formation of robust ohmic contacts between the embedded pre-pattern Pt electrodes and PtSe₂. This ohmic behavior persists across the full temperature range for high V_{BG} (Supplementary Fig. 4a). For $V_{BG} < 7.5$ V the contacts deviate from ohmic behavior at $T = 1.5$ K (Supplementary Fig. 4b); however, at lower gate voltages such as $V_{BG} = 3$ V linear $I - V_{SD}$ characteristics emerge at elevated temperatures (≥ 30 K) and at room temperature linearity extends over a broader range of V_{BG} values (Supplementary Fig. 4c, d). The contact resistance (R_C), determined by evaluating two- and four-

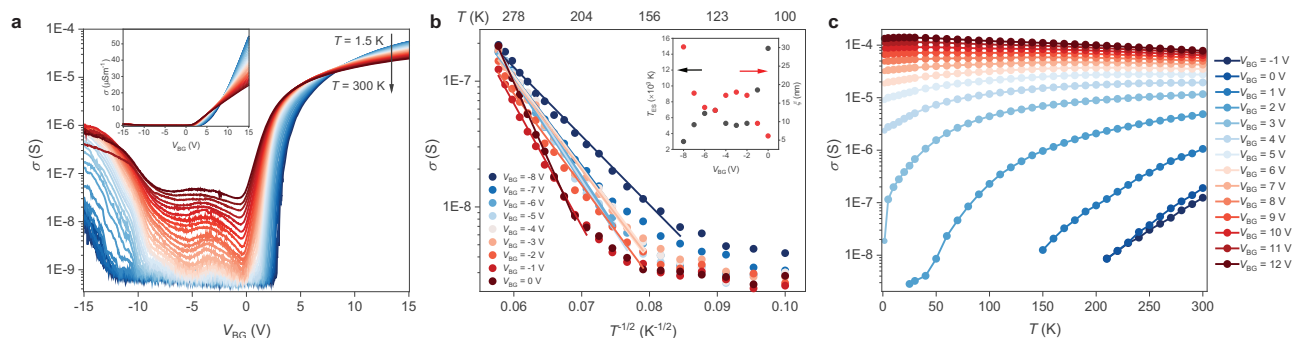


Fig. 3 | Temperature-dependent conductivity measurements across the metal-insulator transition (MIT) in 5 L PtSe₂. **a** σ as a function of backgate voltage for temperatures ranging from 1.5 K to 300 K, plotted on a semilog-scale to magnify the unconventional OFF state conduction. Inset: same plot on a linear scale for comparison. **b** σ plotted against $T^{-1/2}$ on a semilog-scale for various V_{BG} in the OFF state,

solid lines representing fittings to the Efros Shklovskii variable-range hopping model (ES-VRH). The inset shows the extracted localization length and characteristic temperature modulation with V_{BG} . **c** Temperature dependence of conductivity at different V_{BG} values. A MIT is evident, with σ showing distinct insulating and metallic behavior.

terminal measurements and their modulation against temperature, is shown in Fig. 2f. Notably, at $V_{BG} = 15$ V, R_C reaches a minimum value of 1.31 k Ω μ m at $T = 1.5$ K, and increases to 2.4 k Ω μ m at $T = 300$ K. This increase in contact resistance with temperature at high gating further supports the ohmic contact behavior observed^{19–21}. Conversely, the I – V_{SD} characteristics at $T = 1.5$ K for the 4 L device, shown in the inset of Fig. 2c, display slight non-linearity at low drain bias, which we attribute to the larger bandgap inhibiting charge injection. Figure 2d presents line cuts of the source-drain current dependence on V_{BG} for both the 4 L and 5 L devices, extracted from their respective 2D color maps. A subthreshold swing of 135 mV/dec, approaching the theoretical limit of 60 mV/dec, indicates sharp turn-ON/OFF behavior for the 5 L device, along with an ON/OFF ratio of 10^5 (Supplementary Fig. 5). Field-effect mobility is extracted from the linear region of the transfer curves under $V_{SD} = 0.1$ V, using the expression $\mu_{FE} = dl_{SD}/dV_{BG} \cdot L/(WC_iV_{SD})$, where $L = 1$ μ m is the channel length, $W = 4$ μ m is the channel width and C_i is the capacitance per unit area. The capacitance is calculated as $C_i = \epsilon_r \epsilon_0 / (t_{SiO_2} + t_{hBN})$ where $\epsilon_r \sim 3.9$ is the dielectric constant of SiO₂ and hBN, and $(t_{SiO_2} + t_{hBN}) \sim 330$ nm is the combined thickness of the dielectric layers. The two-terminal electron field-effect mobilities of ~ 650 cm²V⁻¹s⁻¹ and ~ 412 cm²V⁻¹s⁻¹ were obtained for the 5 L and 4 L devices, respectively, representing the highest values recorded for PtSe₂-based FETs at these thicknesses^{13,15}. We note that the Au-top-contacted test sample on SiO₂ (Supplementary Note 1) shows significantly lower mobilities, underscoring the importance of contact quality and hBN encapsulation on device performance.

We further examine the mobility of devices by probing in a four-terminal (4-t) geometry; Fig. 2e conveys the evolution of the 4-t and 2-t mobility of the 5 L device as a function of temperature. The highest 4-t mobility is found to be ~ 1640 cm²V⁻¹s⁻¹ at $T = 1.5$ K for the 5 L device and ~ 1480 cm²V⁻¹s⁻¹ at $T = 20$ K for the 4 L device (Fig. 2e inset). As the temperature reduces from 300 K to ~ 100 K, there is a steep increase in the mobility in both measurement geometries, which delineates that mobility is limited by phonon scattering. The temperature dependence in this regime follows the power law, $\mu \sim T^\gamma$, where γ depends on electron-phonon scattering and is extracted to be ~ 0.9 – 0.97 and ~ 1.25 for 5 L and 4 L devices, respectively. Calculations for MoS₂²², suggest that $\gamma \sim 1$, as observed in the 5 L device, is indicative of scattering by acoustic phonons, while $\gamma > 1$, observed in the 4 L device, corresponds to scattering by optical phonons. Additionally, the γ values obtained for our PtSe₂ devices are comparable to those reported for high-quality, fully encapsulated MoS₂¹⁹ and WSe₂⁸, reinforcing the quality of our samples. The mobility continues to increase more gradually as the temperature is lowered further until ~ 50 K, where it begins to saturate before decreasing as the temperature reaches 1.5 K. This behavior suggests that mobility is increasingly limited by charge impurity

scattering at lower temperatures. We will later discuss how the onset of charge-impurity-dominated scattering introduces disorder, leading to a crossover from an interaction-dominated transport regime to percolation-like transport.

We next investigate the temperature-dependent charge transport characteristics of the 5 L device. Figure 3a presents the conductivity as function of V_{BG} on a semilog-scale, measured over the temperature range 1.5–300 K under a constant $V_{SD} = 0.1$ V. Firstly, for gate voltages between -8 V to -15 V, we observe that the device exhibits p-type semiconducting behavior. This is expected given the small size of the bandgap, allowing for the gate voltage to tune the Fermi level into both the conduction and valence bands. Secondly, we detect the crossover of all conductivity curves acquired at different temperatures around $V_{BG} = 7$ V; this is suggestive of a metal-insulator transition. Below this point, the device performs as a classical semiconductor with conductivity decreasing as the temperature is lowered. However, above the cross-over point the conductivity increases as the temperature is reduced, a characteristic of metallic conduction³.

Another striking feature is the unconventional behavior of the OFF-state. As the temperature rises above ~ 100 K, the OFF-state conductivity increases and follows an oscillatory behavior as the gate voltage is swept. We attribute this behavior to hopping conduction of defect induced localized gap states (Supplementary Note 2), similar to previously reported in MoS₂²³. In Fig. 3b, this OFF-state conduction is fitted to the Efros-Shklovskii²⁴ variable range hopping (ES-VRH) model, which describes an interacting disordered system,

$$\sigma(T) \propto \exp[-(T_{ES}/T)^{1/2}] \quad (1)$$

in this expression T_{ES} is the characteristic temperature and is defined as $T_{ES} = 2.8e^2/\epsilon\epsilon_0\xi k_B$, where $\epsilon \sim 3.5$ is the dielectric constant of PtSe₂²⁵, ξ is the localization length and k_B is the Boltzmann constant. The linear regions of the curves are well-fitted to the above equation and are used to extract T_{ES} , from which ξ is discerned. The gate voltage dependence of ξ is plotted in Fig. 3b (inset), we find that $\xi = 30.18$ nm at $V_{BG} = 0$ V and reduces to $\xi = 6.13$ nm at $V_{BG} = -8$ V. A slight saturation of conductivity is observed below ~ 150 K, which likely arises from the minimum detection limit of our setup, with differences in saturated conductance reflecting varying levels of residual current. The OFF-state conduction is well-described by ES-VRH demonstrating that both disorder and strong electron interactions influence the electronic properties of this system.

The carrier density at which the MIT occurs is referred to as the critical carrier density n_C . In Fig. 3c, the conductivity versus the temperature is displayed, we determine $V_C = 3$ V, which from the capacitance model $n = C_i(V_{BG} - V_{TH})/e$; corresponds to $n_C = 2.75 \times 10^{10}$ cm⁻²

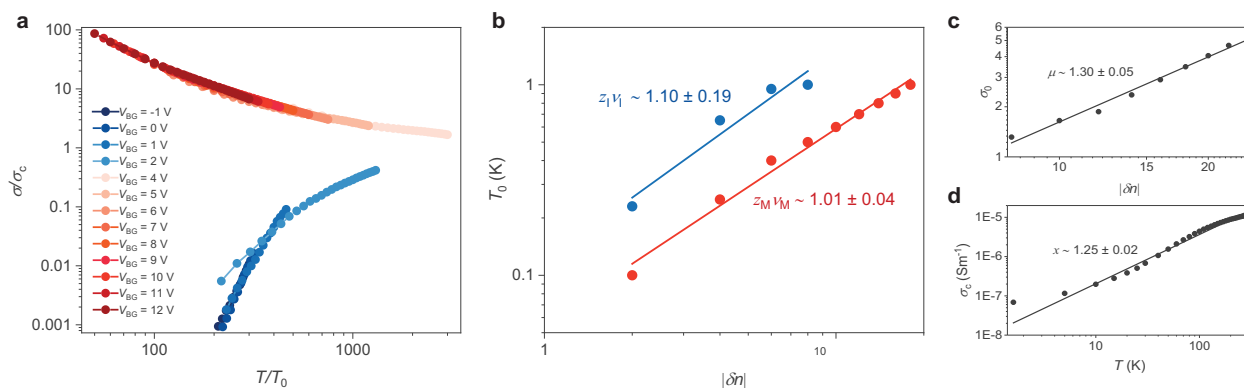


Fig. 4 | Scaling analysis of conductivity in 5 L PtSe₂. **a** Dynamical scaling of normalized conductivity (σ/σ_c) as a function of the rescaled temperature T/T_0 for various V_{BG} near the critical backgate. The scaling is performed for temperatures between 50 K and 300 K revealing two distinct metallic and insulating branches. **b** T_0 plotted against normalized carrier density ($|\delta n|$), on a log-log scale, the red and blue data points representing metallic and insulating sides, respectively. The linear fits extract

critical exponents $z_I \nu_I \approx 1.10 \pm 0.19$ (insulating) and $z_M \nu_M \approx 1.01 \pm 0.04$ (metallic). **c** σ_0 versus $|\delta n|$ on a log-log scale, with the extracted exponent $\mu \approx 1.30 \pm 0.05$ from the linear fit (black line). **d** σ_c as a function of temperature on a log-log scale, with the linear fit (black line) yielding a critical exponent $x \approx 1.25 \pm 0.02$. The consistency of the scaling arguments and the bifurcation of the metallic and insulating branches are consistent with interaction-dominated transport near the transition.

(Supplementary Note 3)^{26–31}. This demonstrates an ultralow critical carrier density in 5 L PtSe₂. Similar V_C was extracted for measurements conducted under zero DC bias and a small AC excitation of 1 mV (Supplementary Fig. 6), indicating no clear influence of potential heating effects on the determination of V_C ³². Theoretical work has shown that n_C is related to mobility and that high mobility due to sample cleanliness results in low n_C ⁷. Accordingly, we attribute the low n_C observed to the high-quality of our samples, demonstrated by the high mobilities achieved and transparency of the contacts. Additionally, the 4 L PtSe₂ device also exhibited a MIT at low carrier densities, with n_C found to be $5.88 \times 10^{10} \text{ cm}^{-2}$ (Supplementary Fig. 7), the difference in n_C compared to the 5 L PtSe₂ suggests that narrowing of the bandgap facilitates achieving lower density MITs. At such low critical densities, carrier interactions are expected to drive the MIT and the strength of the Coulomb interaction between carriers can be characterized by the Wigner-Seitz radius, $r_s = m^*/(m_0 a_B \epsilon \sqrt{\pi n})$ where $m^* = 0.21 m_0$ is the effective mass of PtSe₂¹⁴ and a_B is the Bohr radius. We find for the 5 L device $r_s \sim 20.8$ at n_C , comparable to high-quality 2D electron gas systems³³, placing the system firmly in the strongly interacting regime ($r_s \gg 1$). This strong interaction regime is accompanied by a low Fermi temperature $T_F = E_F/k_B \approx 3.6$ K, where $E_F = \pi \hbar^2 n / m^*$. Furthermore, we characterize the metallic phase as Fermi liquid-like (Supplementary Fig. 8), further indicating the strongly correlated nature of our system⁸.

We now turn our attention to the mechanism underlying the observed MIT. In 2D systems where strong carrier-carrier interactions are present, critical-like transport behavior can emerge near a density separating metallic and insulating regimes. Such behavior is often analyzed using scaling approaches formulated in the context of quantum phase transitions. The temperature dependence of the conductivity in this regime is given by,

$$\sigma(\delta n, T) = \sigma_c(T) F[T/T_0(\delta n)] \quad (2)$$

where σ_c is the critical conductivity used to renormalize σ , F is the universal scaling function, $\delta n = (n - n_C)/n_C$ represents the normalized carrier density and $T_0(\delta n) \propto |\delta n|^z$ with z being the dynamical exponent and ν the correlation length exponent³⁴. In the present work, this scaling analysis is employed as a phenomenological characterization of finite-temperature critical-like transport, rather than as evidence of access to an asymptotic zero-temperature quantum critical regime.

When the temperature is rescaled appropriately for the metallic and insulating regimes, the renormalized conductivity collapses into

two distinct branches. This phenomenon is illustrated in Fig. 4a, where data scaling from 50 to 300 K shows a clear bifurcation of the conductivity into an upper branch for the metallic phase and a lower branch for the insulating phase, near the transition. The successful scaling of the conductivity over a wide temperature range is consistent with interaction-dominated transport near the transition, reflecting the important role of strong carrier-carrier interactions in this regime. However, we note that scaling behavior alone does not uniquely determine the microscopic mechanism, as scaling-like behavior may also arise from other effects⁶. Moreover, the scaling collapse is observed over the temperature range 50–300 K, well above the estimated Fermi temperature at the critical density, and therefore limits the interpretation of the data within a zero-temperature QPT framework. However, when applying the scaling analysis across the full temperature range (1.5 K to 300 K), the data below 50 K does not conform well to the expected scaling (Supplementary Fig. 9). Similar deviations have been reported in MoTe₂/WSe₂ moiré superlattices³⁵ and have been ascribed to the presence of disorder in the sample creating charge inhomogeneity. Interestingly, the temperature range at which the scaling fails, 1.5 to 50 K, coincides with the transition from the phonon-limited regime to the charge impurity scattering regime in the charge transport, as conveyed in Fig. 2e. This correlation suggests that the disorder introduced by charge impurities disrupts the role of strongly correlated electrons in driving the MIT, obscuring any potential quantum critical behavior and preventing the observation of a zero-temperature QPT. Consequently, the disorder reduces the MIT mechanism to a percolation-type transition, which we will discuss in further detail later.

The validity of the scaling analysis can be verified by extracting the critical exponents. In Fig. 4b, T_0 is fitted to the power law $T_0(\delta n) \propto |\delta n|^z$, yielding $z_I \nu_I \approx 1.10 \pm 0.19$ and $z_M \nu_M \approx 1.01 \pm 0.04$, for the insulating and metallic sides, respectively. These values are consistent with those reported for WSe₂²⁷ and Si-MOSFETs^{36,37}. Additionally, the power laws $\sigma_c(n_C, T) \propto T^x$ and $\sigma_0(\delta n, T=0) \propto |\delta n|^\mu$ are used to extract the exponents μ and x , as shown in Fig. 4c, d. The fittings yield values of $x \approx 1.25 \pm 0.02$ and $\mu \approx 1.30 \pm 0.05$, and from standard scaling arguments, we obtain $\mu = (z\nu) \approx 1.32 \pm 0.14$, which is in agreement with the value of μ obtained from the fitting in Fig. 4c. A value of $\mu \sim 1$ –1.5 is also typically associated with low-disorder, strongly interacting systems²⁶. These consistencies further corroborate our observation of interaction-dominated transport and, together with the presence of ES-VRH demonstrated in Fig. 3b and high r_s , provide strong evidence for the significant role of strongly correlated electrons in PtSe₂.

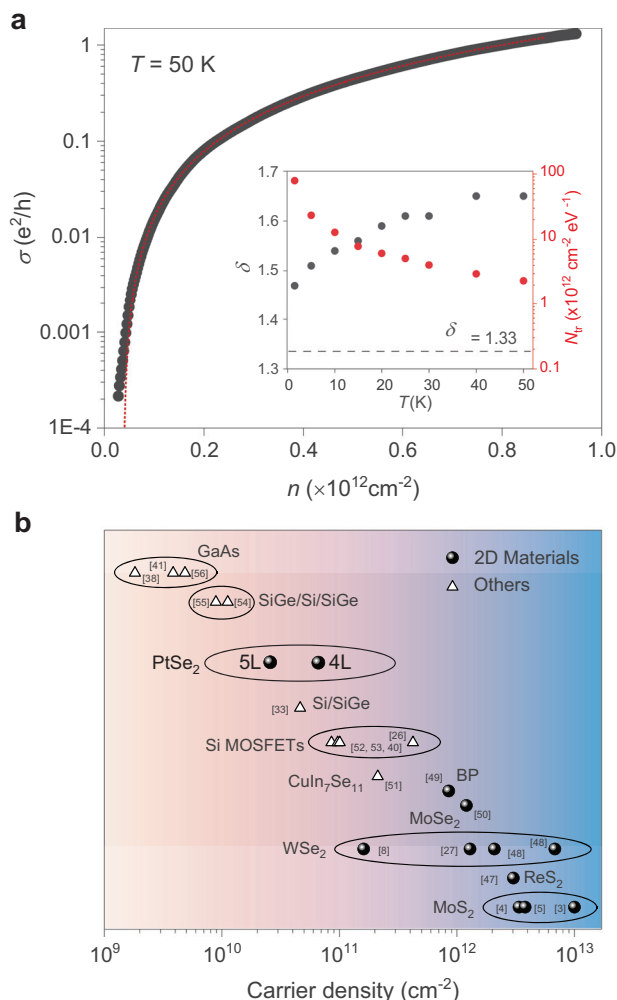


Fig. 5 | Low temperature percolation-type MIT behavior. **a** Conductivity as a function of carrier density at $T = 50$ K, the red dashed line represents the fitting to the percolation model used to extract the percolation exponent, δ . Inset: δ (black) and N_{tr} (red) versus T , showing δ approaching the ideal value 1.33 indicated by the horizontal black dashed line. **b** Comparison of critical carrier density of MIT for the 5 L and 4 L PtSe₂ devices in this study and other 2D materials^{3–5,8,27,48–51} and other 2DEG systems^{26,33,36,38,40,41,52–56}. The shading from white to blue through pink represents increasing critical carrier density.

Finally, we examine the nature of the MIT below 50 K. It follows, in systems undergoing a MIT driven by Coulomb disorder, a percolation-based MIT is expected as the carrier density is modulated, where the conductivity typically follows the relation, $\sigma(n) = A(n - n_{CP})^\delta$, with δ being the percolation exponent which is ≈ 1.33 for 2D systems and n_{CP} the percolation threshold density. Fitting the percolation model to the data at 50 K, as displayed in Fig. 5a, yields $\delta = 1.65$, which is in line with values reported for MoS₂⁵. In fact, we find that as the temperature is reduced towards 1.5 K, δ converges towards 4/3, and this relation is depicted in Fig. 5a (inset), reaching 1.47 at $T = 1.5$ K. The trend indicates that percolation behavior becomes increasingly dominant as the temperature is lowered, suggesting a rise in disorder within the system. Similar converging behavior toward 1.33 as the temperature is reduced has also been observed in n-GaAs systems³⁸. This observation directly correlates with the drastic increase in the effective charge trap density (N_{tr}) as the temperature falls below 50 K, as shown in Fig. 5a (inset). The effective charge-trap density, discerned from the sub-threshold swing³⁹ via the expression $SS = (k_B T/e) \ln(10) (1 + (e^2 N_{tr}/C_i))$, rises from $5.88 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$ at $T = 50$ K to $7.20 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$ at $T = 1.5$ K. This behavior coincides with the observed enhancement in

charge impurity-scattering limiting the mobility as temperature falls below 50 K, as shown in Fig. 2e. The pronounced increase in the relative strength of N_{tr} in this low temperature range overcomes the Coulomb interactions between carriers, forcing a percolation-based MIT driven by disorder rather than strongly correlated electrons.

Discussion

The fabrication of high-quality low band-gap PtSe₂ devices has enabled the observation of a metal-insulator transition at very low carrier densities. As the system tends towards a charge-impurity scattering-dominated transport regime below 50 K, the failure of screening at low carrier densities gives rise to enhanced Coulomb disorder. This leads to a breakdown of finite-temperature scaling and a departure from the interaction-dominated transport regime. In disordered 2D systems, long-range Coulomb scattering introduces spatial inhomogeneities in the potential landscape (“puddles”) which disrupt coherent charge transport and give rise to a classical, percolation-driven transition^{40,41}. The emergence of percolative behavior below 50 K, characterized by the extracted temperature-dependent percolation exponent δ , reflects a fundamental shift in the dominant mechanism governing charge transport in PtSe₂, where charge impurity scattering begins to dominate, gradually overshadowing the electron-electron interactions at higher temperatures. This may explain the observed increase in N_{tr} as temperature is lowered, and charge-impurity scattering strengthens. The MIT can thus be delineated by two distinct regimes: a high temperature, low-disorder, interaction-dominated transport regime and a low temperature, high-disorder percolation-like regime. Although strong electron-electron interactions persist across both regimes, disorder ultimately dominates the MIT mechanism at lower temperatures, consistent with theoretical predictions⁷. This dual behavior of the MIT underscores the complex interplay between electronic interactions and disorder in determining the phase behavior of 2D materials. We further verified that these conclusions are reproduced under zero DC bias conditions (Supplementary Fig. 10), confirming that the observed dual character of the MIT is intrinsic to PtSe₂ and not an artifact of finite-bias heating.

PtSe₂ not only exhibits gate-voltage-driven MIT but also thickness-dependent and temperature-dependent transitions. Notably, the material demonstrates a crossover at a thickness of 6 L, where the MIT behavior transitions (Fig. 1e). Additionally, it shows a pronounced temperature-dependent conductance in the OFF state, as evidenced by the OFF current increasing by two orders of magnitude with rising temperature (Fig. 3a). The salience of 5 L PtSe₂ is underlined by its critical carrier density being almost an order of magnitude lower than other TMD systems, as illustrated in Fig. 5b. This low n_C value approaches those seen in ultrahigh-mobility GaAs MOSFETs and SiGe/Si/SiGe quantum wells, while the increase in n_C for identically fabricated 4 L PtSe₂ devices indicates a dependence on band gap. This underscores the advantage of multilayer PtSe₂’s narrower bandgap in accessing the low-density, strongly interacting regime where quantum phase transitions emerge—conditions that are challenging to reach in wider-bandgap TMDs such as MoS₂ or WSe₂.

Together, these characteristics establish PtSe₂ as a versatile material system, enabling controlled manipulation of transition mechanisms through variations in temperature, carrier density, and disorder. Furthermore, our achieved high mobilities and low contact resistance in PtSe₂, along with advancements in its large-scale growth⁴², emphasize its promise for future electronic device applications.

Methods

Growth of crystals

For the synthesis of PtSe₂, platinum powder (99.99%, SurePure, USA) and selenium granules (99.9999%, 2–4 mm, Wuhan Xinrong New Materials Co., USA) were used. The synthesis was carried out using a self-flux growth method in a quartz glass ampoule. Platinum and

selenium, in amounts sufficient to produce 3 g of PtSe₂, along with 2 at.% excess selenium to ensure stoichiometric balance, were placed in the ampoule (dimensions: 100 × 25 mm, 3 mm wall thickness). The ampoule was sealed under high vacuum (below 1×10^{-3} Pa, achieved using an oil diffusion pump and liquid nitrogen cold trap) and sealed with an oxygen-hydrogen welding torch. The ampoule was placed inside a corundum protective crucible (wall thickness >5 mm) within a muffle furnace equipped with SiC heating elements. The furnace was first heated to 800 °C at a rate of 1 °C/min. After maintaining this temperature for 25 h, the temperature was raised to 1280 °C over 4 h, then reduced to 1230 °C over 120 min, followed by a further reduction to 1000 °C over the next 120 min. The sample was left to cool to room temperature overnight. After cooling, the ampoule was opened in an argon-filled glovebox, and crystals were exfoliated from the resulting bulk material.

BN single crystals were obtained by using the temperature gradient method under high pressure and high temperature. Typical growth conditions were 3 GPa and 1500 °C for 120 h. A solvent of the Ba–BN system was used to obtain high-purity crystals. The source of BN crystals was heat treated so as to reduce oxygen impurities at 2000 °C under a nitrogen atmosphere. The recovered BN crystals were treated with strong acid (hot aqua regia) to remove residual solvent and washed with diluted water to supply them for the exfoliation process.

Sample preparation

Bottom hBN flakes (~30 nm) were exfoliated onto heavily doped 300 nm Si/SiO₂ substrates, which served as global back gates. A bilayer PMMA resist was first spin-coated to produce an undercut profile for lithography. Hall-bar contacts were defined by electron-beam lithography, and the exposed hBN was etched by ~15 nm using CHF₃/O₂ deep reactive ion etching (40/4 sccm, 30 W). Ti/Pt (2/14 nm) pre-patterned electrodes were deposited by electron-beam evaporation, after which the substrates were annealed at 340 °C for 6 h in high vacuum ($\sim 1 \times 10^{-6}$ mbar). A top hBN/PtSe₂ heterostructure prepared by polycarbonate-based dry transfer was then laminated onto the pre-patterned contacts. Ti/Au (2/100 nm) metal leads were subsequently added in a second lithography step, followed by a final anneal under identical conditions. The wafers used are SSP PRIME GRADE Si wafers sourced from NOVA Electronic Materials LLC and have a custom resistivity specification ranging from 0.005 to 0.02 Ωcm. These wafers are heavily doped and highly conductive, suitable for FET devices.

Measurement

Cryogenic measurements were performed in a closed-cycle Oxford Instruments cryomagnetic system with a base temperature of ~1.5 K. During two-terminal charge transport measurements, drain currents were measured using a Keithley Sourcemeter 2450, and a Keithley Sourcemeter 2400 was used to apply gate bias through the SiO₂ gate dielectric. Four-terminal charge transport measurements were performed by using a lock-in amplifier (Stanford Research SR830) at very low-frequency (~13 Hz).

First-principles calculations

Density functional theory calculations were performed using the Vienna Ab-initio Simulation Package (VASP)⁴³ utilizing plane-wave basis set and the projector-augmented wave (PAW) method^{44,45}. Band structure calculations utilized the HSE06 hybrid exchange-correlation functional⁴⁶. An energy cutoff of 250 eV, and a $9 \times 9 \times 1$ uniform sampling of the Brillouin zone were used in the self-consistent field procedure. Periodic images of PtSe₂ multilayers were separated by a 20 Å thick vacuum region. The atomic coordinates of the optimized computational models are provided as a separate Supplementary Data file.

Data availability

The data supporting the findings of this study are available within the paper and its Supplementary files. All the data that support the findings of this study are available from the corresponding author upon request.

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Author contributions

A.A. designed and coordinated the work. A.A.M. fabricated the samples. A.A.M. and Th.T. performed transport measurements. K.W. and Ta.T. grew the hBN crystals. J.F. and O.V.Y. devised models and performed first-principles calculations. O.C. and Z.S. grew the PtSe₂ crystals. A.A.M., Th.T. and A.A. analyzed the results and A.A.M. and A.A. wrote the manuscript with inputs from all authors.

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Competing interests

The authors declare no competing interests.

Additional information

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